XG-440 Firmware Release Notes: v3.14.2 → v3.15.0

Model: VFIP-RN-XG440-2025 | Category: Technical | Package: Firmware & Operational Documentation for RAG use

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1. Executive Summary

This document bundles the release notes, parameter diffs, protective trip definitions, rollout tickets, O&M incident report, runbooks, and settlement guidance for inverter model XG-440 (VFIP-RN-XG440-2025) covering firmware revisions v3.14.2 through v3.15.0. The staged rollout occurred 2025-11-20 → 2025-11-21 with a subsequent hotfix applied 2025-12-08. Key changes include grid synchronization logic changes, new protective trip codes (TRIP-31, TRIP-44), modified default thresholds for low-voltage ride-through, and telemetry additions to support SCADA alarm mapping.

> Intended audience: Field O&M engineers, control engineers, SCADA integrators, asset managers, and RAG systems indexing this pack for automated assistance.

2. Scope & Applicability

Firmware scope: all XG-440 inverters running vendor firmware v3.14.2 upgraded to v3.15.0. Applies to serial ranges and plant scopes noted in Section 5. Includes software-level changes (no hardware changes).

Model	Firmware Range	Applicable Serial Ranges	Notes

VFIP-RN-XG440-2025	v3.14.2 → v3.15.0	SN: XG440-000100 → XG440-001600	Staged rollout; hotfix for TRIP-44 timing

Prerequisites: backup of device configuration, SCADA telemetry mapping verification, site-level rollback plan including spare inverters or quick re-dispatch capacity. See Section 6 for rollback procedure.

3. Version-by-Version Parameter Diff

This section lists concrete parameter names and changes, default values (old → new), and rationale.

3.1 v3.14.3 (interim patch)

Parameter	Field Identifier	Old Default	New Default	Rationale
Low Voltage Ride Through Timeout	ride_through_timeout_s	10	12	Increased margin to comply with recent grid-code tests; reduces nuisance re-trips for transient sags.
Grid Sync Mode	grid_sync_mode	"phase_lock"	"adaptive_phase_lock"	Improved synchronization algorithm under weak-grid conditions.
Protect Threshold (per-unit)	protect_threshold_pu	0.85	0.83	Reduce risk of unnecessary shedding on voltage sags.

Notes

The v3.14.3 patch was limited and aimed to prepare devices for the broader logic changes in v3.15.0. No new SCADA telemetry added in this patch.

3.2 v3.15.0 (major release)

v3.15.0 introduces new protective trips, telemetry flags, configuration fields, and modified defaults.

Parameter	Field Identifier	Old Default	New Default	Rationale
Overcurrent Threshold	oc_protect_threshold_A	520	510	Lowered to prevent thermal stress on newly observed cable runs.

Reactive Power Ramp Limit	q_ramp_limit_pct_per_s	10	8	Smoother reactive response required by some DNOs.
New Telemetry: Inverter Availability Flag	inverter_availability_flag	—	boolean	Binary flag for SCADA to indicate firmware-determined availability state.
SCADA Alarm Code Field	scada_alarm_code	—	integer	Standardized alarm codes to map to CMMS (see Section 8).
DC Overvoltage Trip Threshold	dc_ov_threshold_V	1100	1120	Adjusted after field reports of DC overvoltage spikes; reduces false trips while protecting components.

Behavioral Changes

- Adaptive synchronization algorithm may vary phase-lock behavior; grid_sync_mode now supports "adaptive_phase_lock" with fallback to "phase_lock".
- Telemetry now includes scada_alarm_code and inverter_availability_flag for each unit (see Section 11).
- Protective-trip hysteresis behavior changed — see TRIP-31 and TRIP-44 descriptions in Section 4.

4. Protective Trip Codes & Details

Includes symptoms, root causes, step-by-step resolution, error codes, and prevention tips for each trip code introduced or modified.

TRIP-31: Grid Frequency Out-of-Limits (Adaptive)

Code	scada_alarm_code	Symptom	Root Cause	Action
TRIP-31	3101	Inverter disconnects with brief ramp-down then OFF	Grid frequency excursions beyond adaptive thresholds; firmware now applies dynamic thresholds based on grid_sync_mode.	Step 1: Verify scada_alarm_code == 3101 and frequency telemetry. Step 2: Confirm ride_through_timeout_s was exceeded. Step 3: If false positive, adjust protect_threshold_pu or revert grid_sync_mode to "phase_lock". Step 4: If persistent, schedule CMMS corrective work order (template: CMMS-FW-TRIP-31).

Troubleshooting flow (TRIP-31):

1. Check SCADA frequency trace ±60s around event; annotate timestamped samples.
2. Confirm inverter_availability_flag toggled to false.
3. Query device logs via vendor API for event:
   `GET /api/v1/inverters/{serial}/events?code=3101&start=2025-11-20T00`
4. Apply mitigation: increase ride_through_timeout_s by +3s for the cohort and monitor.

TRIP-44: DC Overvoltage Hysteresis Trip (New)

Code	scada_alarm_code	Symptom	Root Cause	Action
TRIP-44	4403	Instantaneous inverter OFF with DC_OV alarm in vendor logs	Transient DC spike surpassing dc_ov_threshold_V with narrow hysteresis window in v3.15.0 default.	Step 1: Confirm scada_alarm_code == 4403 and check dc_voltage trace. Step 2: If spike is transient (<200ms) and repeated, apply hotfix (2025-12-08) or increase dc_ov_threshold_V by +10-20V and widen hysteresis setting dc_ov_hysteresis_V. Step 3: If hardware issue suspected (PV string isolation fault), dispatch field crew.

Resolution Steps and Preventive Measures

- For transient DC spikes: implement firmware hotfix v3.15.1 (hotfix deployed 2025-12-08) or adjust dc_ov_hysteresis_V = 6 (default was 3).
- For frequency excursions (TRIP-31): tune ride_through_timeout_s and monitor network frequency stability; coordinate with TSO if persistent.
- Record and tag every trip event in CMMS with vendor event export for post-analysis: include device serial, firmware version, scada_alarm_code, timestamps, and pre/post-config snapshots.

5. Change Ticket Summaries (Rollout & Hotfix)

Extracted from vendor portal change tickets. Includes staged rollout schedule, serial ranges, and scope by plant.

Rollout: 2025-11-20 → 2025-11-21 (Ticket: VENDOR-CHG-2025-110)

Ticket ID	Schedule	Scope (Plants)	Serial Ranges	Notes

VENDOR-CHG-2025-110	2025-11-20T02:00Z → 2025-11-21T18:00Z	Plant A (ND-001), Plant B (ND-002), Plant C (ND-003)	XG440-000100 → XG440-000600 (Plant A)	Staged in 50-unit batches; pre-upgrade config snapshot mandatory.

Rollout action items: take configuration snapshot (API export), verify SCADA mapping, ensure field crew on-call for second shift. Staged timestamps for each batch were recorded in ticket comments (see example timeline below).

Sample Staged Timestamps (from ticket comments)

```
2025-11-20T02:15Z - Batch 1 (XG440-000100..000149) started
2025-11-20T03:02Z - Batch 1 completed, validation passed
2025-11-20T05:30Z - Batch 2 (XG440-000150..000199) started
2025-11-20T06:10Z - Batch 2 completed, 2 units reported TRIP-31 events
(investigating)
2025-11-21T09:00Z - Batch N completed
```

Hotfix: 2025-12-08 (Ticket: VENDOR-HOTFIX-2025-122)

Ticket ID	Date	Scope	Serial Ranges	Purpose
VENDOR-HOTFIX-2025-122	2025-12-08T01:30Z	All units that experienced TRIP-44 events during rollout	XG440-000200 → XG440-001600 (selective)	Adjust DC overvoltage hysteresis and apply stability patch

Hotfix was applied selectively based on ticket analysis (error code 4403 correlation). Validation included 72-hour monitoring window for repeated TRIP-44 events.

6. Rollback Conditions & Procedures

Rollback should be executed only when defined conditions are met. Follow the step-by-step procedure below; emphasize preservation of logs and SCADA data for root cause analysis.

Rollback Conditions (any one of the following)

- More than 5% of a batch experience protective trips (TRIP-31 or TRIP-44) within 24 hours post-upgrade.
- Inverter_availability_flag shows false for >30% of units at a plant for >2 hours.
- Unacceptable curtailment resulting in projected imbalance charges exceeding threshold (see Section 9).
- Critical hardware faults identified (PV isolation, thermal events).

Rollback Prerequisites

1. Ensure you have the pre-upgrade firmware image (v3.14.2) stored and verifiably checksum-matched.
2. Export configuration snapshot:
   `GET /api/v1/inverters/{serial}/config?version=pre-upgrade`

3. Ensure remote or local access to perform downgrade; confirm maintenance window and notify grid operator if required.

Step-by-Step Rollback Procedure

1. Place inverter into maintenance mode via SCADA: set scada_inhibit = true.
2. Disable automated fleet upgrades for the affected serial range in vendor portal.
3. Push firmware v3.14.2 package:

```
POST /api/v1/inverters/{serial}/firmware
{
  "fw_image": "v3.14.2.bin",
  "action": "downgrade",
  "preserve_config": true
}
```

4. Monitor progress: firmware_state should transition: downloading → flashing → reboot → operational.
5. After reboot, verify:
   - inverter_availability_flag == true
   - scada_alarm_code == 0 (no active alarms)
   - telemetry match pre-upgrade baselines
6. If rollback fails on >2 devices, escalate to vendor support with attached logs and SCADA traces (see Vendor Support Packet template in Appendix A).

Verification Checklist

- Configuration keys preserved: protect_threshold_pu, ride_through_timeout_s, grid_sync_mode.
- No new spurious protective trips within 24 hours.
- Documented change in CMMS and vendor portal closure of the rollback ticket.

7. O&M Incident Report & Postmortem

Chronological, factual account of the incident associated with the rollout and actions taken.

Timeline (Day-by-Day)

Date UTC	Event	Action
2025-11-20T02:15Z	Rollout batch 1 started	Deployment began; snapshot saved
2025-11-20T06:10Z	2 units reported TRIP-31	Triggered initial investigation; increased ride_through_timeout_s on subsequent batches
2025-11-20T18:45Z	Multiple DC overvoltage events observed	Paused rollout pending analysis; SCADA alarms 4403 logged
2025-11-21T10:30Z	Rollback of last 3 batches initiated	Rollback executed for XG440-000350..000449

2025-12-08T01:30Z	Hotfix applied (VENDOR-HOTFIX-2025-122)	Hysteresis widened; monitoring window started

Root Cause Analysis (summary)

Primary cause: v3.15.0 introduced a narrower DC-overvoltage hysteresis (dc_ov_hysteresis_V default reduced), which caused transient DC spikes on certain PV string topologies to produce TRIP-44 events. Secondary contributor: grid frequency dynamics led to TRIP-31 events on weak-grid sites when ride_through_timeout_s default was insufficient for observed excursions.

Corrective & Preventive Actions

- Hotfix v3.15.1 (applied 2025-12-08) increased dc_ov_hysteresis_V default to 6V and added protective logging.
- Updated pre-deployment checklist to include validation of site DC voltage waveforms and per-string spike characterization.
- Enhanced telemetry to include pre-trip 1-second raw DC samples for every TRIP-44 occurrence.
- Changed rollout governance to require a 48-hour verification window for cohorts larger than 200 units.

Curtailment Window Description

During the incident, several inverters transitioned to not-available status causing automatic curtailment actions on plant-level control. A curtailment window is defined as a contiguous period when effective capacity is reduced by >5% of plant AC capacity for longer than 5 minutes.

Documented windows and estimated lost production are included in site-level reports and were used to calculate imbalance exposures (see Section 9).

Validation Steps After Mitigation

1. Monitor DC voltage trace at 200 ms resolution for 72 hours.
2. Confirm no TRIP-44 events reoccur across patched cohort.
3. Check inverter_availability_flag and scada_alarm_code for 48-hour stability.
4. Run post-upgrade performance test: 2-hour frequency disturbance simulation if allowed by DNO, otherwise monitor live grid behavior.

8. Dispatch Runbooks: SCADA → CMMS Mapping

Standardized mapping of SCADA alarms (scada_alarm_code) to CMMS corrective work order templates and priority routing.

scada_alarm_code	Alias	Description	CMMS Template	Priority
3101	TRIP-31	Grid frequency trip	CMMS-FW-TRIP-31	P1 (Investigate, 2h)

4403	TRIP-44	DC overvoltage trip	CMMS-FW-TRIP-44	P1 (Field dispatch if hardware suspected)
2000	INV-UNRESP	Inverter unresponsive to SCADA ping	CMMS-HW-INV-PING	P2 (24h)
0	NORMAL	No active alarm	—	—

CMMS Corrective Work Order Template: CMMS-FW-TRIP-44 (Example)

Title: Investigate TRIP-44 DC Overvoltage Events
Priority: P1
Description: Multiple TRIP-44 events reported from inverter(s) {serial_range}. Attach vendor event logs and 200ms-resolution DC voltage traces for the 60s pre/post event window. Verify PV string isolation and measure DC ripple.
Tasks:

1. Collect & upload logs (vendor API) and SCADA traces.
2. Measure DC string isolation resistance (IR test).
3. If hardware suspected, replace affected string combiner or PV module and retest.
4. If configuration suspected, coordinate with control engineer to apply config change or schedule rollback.

Automated CMMS Triggering Rules

1. If scada_alarm_code == 4403 and occurrence_count >= 2 within 24h → auto-create CMMS-FW-TRIP-44.
2. If inverter_availability_flag == false for >60 minutes and scada_alarm_code != 0 → create CMMS-HW-INV-PING.

9. Settlement & Imbalance Charge Guidance (ERCOT)

Guidance to estimate penalties, reconciliation fields to collect, and calculation examples for ERCOT settlement exposure due to curtailment windows during firmware incident.

Relevant Data Fields to Reconcile

- Settlement interval timestamps (TIU intervals), resolution 5 minutes
- Aggregated actual generation reported (MW) per interval
- Scheduled generation/offer (MW) per interval
- SCADA alarm timestamps for curtailment window start/end
- Inverter availability per interval (inverter_availability_flag aggregated)

Penalty Calculation (ERCOT) — Conceptual

Imbalance charges are calculated as difference between scheduled (or obligated) and actual delivered energy, multiplied by the prevailing settlement price (LBMP/RTM). Simple exposure estimate:

```
For each 5-min interval i:
```

```
Loss_i_MWh = max(0, Scheduled_i_MW - Actual_i_MW) * (5/60)
Exposure_i_$ = Loss_i_MWh * SettlementPrice_i_$/MWh
TotalExposure_$ = Σ Exposure_i_$ over curtailment window
```

Worked Example

Plant scheduled = 50 MW for interval; actual delivered due to firmware trip = 40 MW; settlement price = $150/MWh.

```
Loss = (50 - 40) * (5/60) = 10 * 0.083333 = 0.83333 MWh
Exposure = 0.83333 * 150 = $125.00 per interval
```

Recommended Reconciliation Fields to Submit to ERCOT and Internal Audit

- TIU interval list with scheduled vs actual values (CSV format)
- SCADA event list mapping intervals to curtailment windows (include scada_alarm_code)
- Firmware version per inverter during each interval
- Corrective action logs and timestamps for transparency

Mitigation Tips to Reduce Settlement Exposure

- Immediately notify market operator of forced outage events to attempt good-faith adjustments.
- Provide evidence packet: pre/post logs, remote commands, and mitigation actions.
- For future rollouts, schedule low-risk windows with reduced exposure (e.g., off-peak periods).

10. Configuration Examples & JSON Snippets

Representative configuration payloads for vendor API usage, recommended settings after v3.15.0 or hotfix, and sample queries to extract logs used in RAG indexing.

Recommended Post-Hotfix Default Configuration (JSON)

```
{
  "firmware_version": "v3.15.1",
  "protect_threshold_pu": 0.83,
  "ride_through_timeout_s": 12,
  "grid_sync_mode": "adaptive_phase_lock",
  "dc_ov_threshold_V": 1120,
  "dc_ov_hysteresis_V": 6,
  "oc_protect_threshold_A": 510,
  "q_ramp_limit_pct_per_s": 8,
  "scada_alarm_code_reporting": true,
  "inverter_availability_flag_reporting": true
}
```

Sample API Query: Export Events for RAG Indexing

```
GET /api/v1/inverters/events?
serial=XG440-000201&start=2025-11-20T00:00Z&end=2025-12-09T00:00Z&fields=timestamp,coc
```

Sample Device Configuration Patch (Selective)

```
PATCH /api/v1/inverters/{serial}/config
Content-Type: application/json

{
  "patch": [
    { "op": "replace", "path": "/ride_through_timeout_s", "value": 15 },
    { "op": "replace", "path": "/dc_ov_hysteresis_V", "value": 6 }
  ]
}
```

Log Export Format (Recommended for RAG ingestion)

Provide newline-delimited JSON (NDJSON) containing: timestamp, serial, firmware_version, scada_alarm_code, inverter_availability_flag, event_type, pre_samples (200ms-resolution array for DC voltage), and post_samples.

```
{
  "timestamp": "2025-11-20T06:09:58.200Z",
  "serial": "XG440-000213",
  "firmware_version": "v3.15.0",
  "scada_alarm_code": 4403,
  "inverter_availability_flag": false,
  "event_type": "DC_OV_TRIP",
  "pre_samples": [1118, 1119, 1121, 1124, 1130],
  "post_samples": [1122, 1135, 1148, 1120]
}
```

11. Recommended Monitoring Metrics & Telemetry Flags

List of telemetry fields to observe after upgrade for early detection and inclusion in RAG datasets.

Field	Type	Sampling Rate	Alert Threshold	Notes
scada_alarm_code	integer	1s	!= 0	Primary alarm mapping for CMMS runbook
inverter_availability_flag	boolean	1s	false for >300s	Indicates firmware-determined operational state
dc_voltage (string-level)	float (V)	200ms (on event) / 1s otherwise	dc_ov_threshold_V + hysteresis	High-resolution capture required for TRIP-44 analysis
frequency	float (Hz)	1s	<49.5 or >60.5 (local policy)	Used to diagnose TRIP-31 incidents
ac_power	float (kW)	5s	refers to expected schedule deviation	For imbalance calculations

event_log_snapshot	ndjson	on trip	—	Attach pre/post samples and config

Alerting Recommendations

- Automated alert when scada_alarm_code in (3101, 4403) with a count >=2 in 24h.
- Alert if inverter_availability_flag false for >60 minutes for >10% of units at plant.
- Store and index event payloads in RAG datastore with vector embeddings for quick root-cause retrieval.

12. Known Issues & Workarounds

List of known issues discovered during rollout and recommended workarounds until permanent fixes are in place.

Issue: Intermittent TRIP-44 on certain string topologies

Workaround: Increase dc_ov_hysteresis_V to 6 and apply hotfix v3.15.1. Avoid rollback unless hardware fault suspected.

Issue: Higher-than-expected reactive ramp causing dispatch oscillations

Workaround: Set q_ramp_limit_pct_per_s to 5 for plants with high reactive sensitivity pending field testing.

Issue: scada_alarm_code reporting latency

Workaround: Increase SCADA polling rate to 1s for alarm-critical fields; implement event-driven webhook from vendor when available.

Issue: False-positive TRIP-31 under microgrid islanding tests

Workaround: During planned islanding tests, set grid_sync_mode = "phase_lock" to avoid adaptive thresholds causing trips; revert after test.

13. Appendices

Appendix A — Parameter Index (alphabetical)

Parameter	Field Identifier	Description
DC OV Hysteresis	dc_ov_hysteresis_V	Voltage hysteresis used to suppress transient DC spikes
DC OV Threshold	dc_ov_threshold_V	DC overvoltage trip level
Grid Sync Mode	grid_sync_mode	Synchronization algorithm selection
Inverter Availability	inverter_availability_flag	Binary flag reported to SCADA
Protect Threshold (pu)	protect_threshold_pu	Per-unit protect threshold for LVRT

Ride Through Timeout	ride_through_timeout_s	Seconds allowed for ride-through events
SCADA Alarm Code	scada_alarm_code	Integer code mapping to vendor-defined alarms

Appendix B — Error Codes & Recommended Resolution Matrix

Error Code	Meaning	Immediate Action	Resolution
3101	TRIP-31: Frequency OOL	Check frequency trace; set ride_through_timeout_s to +3s for cohort	Coordinate with TSO; tune adaptive thresholds
4403	TRIP-44: DC OV	Inspect DC trace; increase dc_ov_hysteresis_V; apply hotfix	Replace PV hardware if persistent; widen hysteresis
2000	INV-UNRESP	Ping device; verify network connectivity	Power cycle inverter; open CMMS-HW-INV-PING if persists

Appendix C — Vendor Support Packet Template

Required attachments:

1. Device serial and firmware version list
2. Pre-upgrade and post-upgrade configuration snapshots (JSON)
3. NDJSON event export including pre/post samples
4. SCADA traces for voltage, frequency, and AC power at 1s resolution (and higher on event)
5. CMMS ticket numbers & actions taken

Appendix D — Contact & Escalation

Vendor support: vendor-support@example.com; Phone: +1-800-VENDOR1; Escalation to engineering required if rollback fails on >5% of batch.

Document Control
Version: v1.0 (compiled 2025-12-10)
Author: O&M Technical Documentation Team
Reviewers: Vendor Firmware Team, Grid Compliance Team
Distribution
Controlled distribution to: O&M, SCADA, Fleet Operations, Asset Management, Market Settlement.
For RAG ingestion: index sections and NDJSON logs; map scada_alarm_code and inverter_availability_flag as primary retrieval keys.